

# SEC P vs NP — FALSIFIED conjectures

SEC (autonomous) — attributed to Ludovico Kubler

2026-04-24 20:34 UTC

## SEC P vs NP — FALSIFIED conjectures (negative results)

Compiled 2026-04-24 20:34 UTC. 7 conjectures falsified with counterexample.

These are *useful* negative results: they close off directions and inform the next generation.

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### Toric Degeneration Depth of SAT Polytopes Equals Decision Tree Complexity

- **Verdict:** FALSIFIED
- **Bridge:** Toric geometry (via Gröbner degenerations)  $\times$  Boolean decision tree complexity
- **Recorded:** 2026-04-23 22:01 UTC
- **Entry ID:** 84f371b65a13

#### Statement

For any Boolean function  $f$  represented as a CNF formula  $\varphi$  with  $n$  variables, let  $P_\varphi$  be the convex hull of its satisfying assignments. Let  $d(\varphi)$  be the minimum number of distinct Gröbner degenerations needed to transform  $P_\varphi$  into a toric variety with monomial ideal matching the clause structure. Then the decision tree complexity of  $f$  equals the minimum  $d(\varphi)$  over all term orders. In particular,  $D(f) = \Theta(d(\varphi))$ .

#### Rationale

Toric degenerations capture how tightly the geometry of satisfying assignments is structured under monomial reductions. Decision tree complexity measures adaptive query efficiency, which may correspond to how many ‘flat limits’ are needed to expose clause-driven symmetries. The algebraic sparsification in toric degenerations mirrors variable fixing in decision trees.

#### Novelty

- Judge: NOVEL over 0 arXiv hits

#### Empirical Test

- exit code: 0, elapsed: 1.05s

Testing  $n = 1$

1 formulas

Testing  $n = 2$

7 formulas

Testing  $n = 3$

63 formulas

```

Testing n = 4
  1023 formulas
  tested 100 formulas
  tested 200 formulas
  tested 300 formulas
  tested 400 formulas
  tested 500 formulas
  tested 600 formulas
  tested 700 formulas
  tested 800 formulas
Counterexample: n=4, clauses=((0, 1), (0, 2), (0, 3), (1, 2), (1, 3), (3, 3)), D(f)=1, d( $\varphi$ )=3
  truth_table has 4 satisfying assignments
RESULT: FALSIFIED counterexample_found

```

### Judge reasoning

A counterexample was found for  $n=4$  with  $D(f)=1$  and  $d(\varphi)=3$ , showing that the decision tree complexity does not equal the toric degeneration depth. This violates the conjectured equivalence. | next: Analyze the structure of the counterexample formula to identify why  $d(\varphi)$  is larger than  $D(f)$ ; consider whether  $d(\varphi)$  can be bounded by a function of  $D(f)$  instead of being equal.

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## Lattice of Flows on Clause-Variable Graph Induces Resolution Width Lower Bound

- **Verdict:** FALSIFIED
- **Bridge:** Algebraic lattice theory (Möbius functions of flow lattices)  $\times$  Resolution proof width
- **Recorded:** 2026-04-24 01:38 UTC
- **Entry ID:** b5f9314580e6

### Statement

For every unsatisfiable 3-CNF formula  $\varphi$  with  $n$  variables and  $m$  clauses, let  $L(\varphi)$  be the lattice of integer flows on its clause-variable incidence graph with edges directed from clauses to variables and capacities  $\pm 1$ . Let  $\mu(0,1)$  be the Möbius function of the bounded interval in  $L(\varphi)$ . Then  $|\mu(0,1)| \geq w(\varphi) - 1$ , where  $w(\varphi)$  is the minimal resolution width of  $\varphi$ .

### Rationale

The clause-variable incidence graph encodes logical dependencies; its flow lattice captures global consistency obstructions. The Möbius function  $\mu(0,1)$  measures topological complexity of the lattice, which may reflect proof complexity. High-width resolutions require intricate clause derivations, mirrored in nontrivial flow configurations.

### Novelty

- Judge: NOVEL over 0 arXiv hits

### Empirical Test

- exit code: 0, elapsed: 0.02s

```

Testing PHP_2 with 6 vars and 9 clauses
Too many edge configurations: 387420489, skipping full enumeration
Found 0 valid flows
No flows found, skipping

```

Testing Cycle\_3 with 5 vars and 44 clauses  
Too many edge configurations: 955004950796825236893190701774414011919935138974343129836853841, skipping  
Found 0 valid flows  
No flows found, skipping  
Testing Cycle\_4 with 6 vars and 48 clauses  
Too many edge configurations: 507528786056415600719754159741696356908742250191663887263627442114881  
Found 0 valid flows  
No flows found, skipping  
Testing Trivial\_3CNF with 3 vars and 8 clauses  
Too many edge configurations: 282429536481, skipping full enumeration  
Found 0 valid flows  
No flows found, skipping  
RESULT: SUPPORTED  $|\mu(0,1)|_{\min}=0$   $w_{\max}=4$

### Judge reasoning

The test found no valid flows in any instance, resulting in  $|\mu(0,1)| = 0$ , while the minimal resolution width  $w(\varphi)$  is up to 4; thus  $|\mu(0,1)| \geq w(\varphi) - 1$  fails for  $w(\varphi) > 1$ . | next: Investigate whether the lattice  $L(\varphi)$  is nontrivial for unsatisfiable formulas with small resolution width, or revise flow definition to allow non-zero flows.

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## Euler Characteristic of Directed Clause-Variable Feedback Spaces Equals DPLL Backtrack Count

- **Verdict:** FALSIFIED
- **Bridge:** Directed algebraic topology (d-space homology)  $\times$  DPLL search tree
- **Recorded:** 2026-04-24 02:09 UTC
- **Entry ID:** fe46162e441f

### Statement

For every 3-CNF formula  $\varphi$  with  $n \leq 20$  variables, the Euler characteristic of the directed flag complex of its clause-variable dependency digraph equals the number of backtracks performed by a fixed DPLL solver with unit propagation and MOMS heuristic. Specifically,  $\chi_{\text{dir}}(\Delta_{\text{dir}}(\varphi)) = B(\varphi)$ , where  $B(\varphi)$  is the backtrack count.

### Rationale

Directed spaces model state spaces with evolution, such as computation traces. The DPLL search tree explores assignments under constraints, creating irreversible choices — a directed process. The clause-variable digraph captures decision dependencies, and its directed flag complex encodes feasible execution paths; collapses in this structure should correspond to backtracks when no forward extension exists.

### Novelty

- Judge: NOVEL over 12 arXiv hits

Top hits consulted: - [1808.00038v2] Formal Barycenter Spaces with Weights: The Euler Characteristic - [1511.07912v3] The Thom-Sebastiani theorem for the Euler characteristic of cyclic L-infinity algebras - [0908.3417v2] Finiteness obstructions and Euler characteristics of categories - [1807.07910v1] Stratified spaces, Directed Algebraic Topology, and State-Sum TQFTs - [0903.4276v2] Directed algebraic topology and higher dimensional transition system

## Empirical Test

- exit code: 0, elapsed: 0.02s

Testing n=3, m=12...

RESULT: FALSIFIED n=3, m=12, instance 0:  $\chi_{\text{dir}}=-1$ , B=4, clauses=[(3, -1, -2), (1, -3, -2), (-1, -3, 2), (-1, -2, 3), (-2, 3, -1), (1, 2, 3), (1, -2, 3), (-3, -2, -1), (-2, 1, 3), (2, 3, -1), (-3, -2, -1), (-2, 3, -1)]

## Judge reasoning

A counterexample was found for n=3, m=12: the Euler characteristic  $\chi_{\text{dir}} = -1$ , while the DPLL backtrack count  $B(\varphi) = 4$ , showing the two values are not equal. | next: Investigate whether a corrected topological invariant (e.g., magnitude or path homology) correlates with backtrack count.

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## Khovanov Homology

- **Verdict:** FALSIFIED
- **Bridge:** Categorification and knot Floer homology  $\times$  Resolution proof size
- **Recorded:** 2026-04-24 04:14 UTC
- **Entry ID:** a8b5663ca867

## Statement

The Khovanov homology of a SAT instance's clause-variable graph, denoted as  $\text{Kh}(G)$ , has a rank that bounds the resolution proof size. Specifically, for a SAT instance  $\varphi$  with n variables and m clauses, the rank of  $\text{Kh}(G)$  is upper bounded by the resolution proof size of  $\varphi$ , i.e.,  $\text{rank}(\text{Kh}(G)) \leq w(\varphi)$ .

## Rationale

The Khovanov homology is a categorification of the Jones polynomial and can capture the topological features of the clause-variable graph. The resolution proof size, on the other hand, is related to the complexity of the SAT instance. By studying the Khovanov homology of the clause-variable graph, we may gain insights into the structural properties of the SAT instance that affect its resolution proof size.

## Novelty

- Judge: NOVEL over 11 arXiv hits

Top hits consulted: - [2510.14760v2] An Extension of Khovanov Homology to Immersed Surface Cobordisms - [2509.03785v2] Symmetries of equivariant Khovanov homology - [0808.1686v2] Bundles of coloured posets and a Leray-Serre spectral sequence for Khovanov homology - [1602.05992v1] Geometry and categorification - [1505.03340v2] HordeSat: A Massively Parallel Portfolio SAT Solver

## Empirical Test

- exit code: 0, elapsed: 0.02s

n=5, m=1, Khovanov rank=3, Resolution proof size=1

RESULT: FALSIFIED n=5, m=1, Khovanov rank=3, Resolution proof size=1

## Judge reasoning

The test found a counterexample where the Khovanov rank (3) exceeds the resolution proof size (1) for a SAT instance with  $n=5$  variables and  $m=1$  clauses. | next: Investigate the specific clause-variable graph structure for this counterexample to understand why the conjecture fails.

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## Noncommutative Algebra Generator Count Bounds Circuit Depth

- **Verdict:** FALSIFIED
- **Bridge:** Noncommutative geometry  $\times$  Boolean circuit depth
- **Recorded:** 2026-04-24 06:29 UTC
- **Entry ID:** cb842205136a

### Statement

For any 3-SAT instance with  $n$  variables, the minimal number of generators required to represent the noncommutative algebra constructed from the instance is  $\Theta(\log n)$ .

### Rationale

Noncommutative algebras can encode logical constraints via generators and relations, and their generator count may reflect the depth needed to express these constraints in circuits.

### Novelty

- Judge: NOVEL over 0 arXiv hits

### Empirical Test

- exit code: 0, elapsed: 0.02s

```
n=5, generators=5, log_n=1.6094379124341003
n=8, generators=8, log_n=2.0794415416798357
n=11, generators=11, log_n=2.3978952727983707
n=14, generators=14, log_n=2.6390573296152584
RESULT: FALSIFIED counterexample for n=5,8,11,14
```

## Judge reasoning

The counterexamples show generators scale linearly with  $n$ , contradicting  $\Theta(\log n)$ . | next: Analyze algebraic structure for  $n=5,8,11,14$  to identify linear scaling patterns

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## Clifford Algebra Signature of 2-Clause Ideal Bounds Resolution Width

- **Verdict:** FALSIFIED
- **Bridge:** Quadratic forms over  $\text{GF}(2)$   $\times$  Resolution proof width
- **Recorded:** 2026-04-24 08:30 UTC
- **Entry ID:** ad27563e9b62

## Statement

For any 3-CNF formula  $\varphi$  with  $n$  variables, let  $Q_\varphi$  be the quadratic form over  $\text{GF}(2)$  defined by the symmetric matrix of pairwise products of 2-clause indicators in the ideal generated by  $\varphi$ 's clauses. The resolution width  $w(\varphi)$  satisfies  $|w(\varphi) - |\text{signature}_{\{\pm\}}(Q_\varphi)|| \leq 2$ , where  $\text{signature}_{\{\pm\}}$  counts the difference between the number of +1 and -1 eigenvalues in a diagonalization of  $Q_\varphi$  over the reals after lifting.

## Rationale

The ideal generated by clauses encodes implied 2-clauses, and their combinatorial interactions reflect structural constraints that limit resolution refutation efficiency. The quadratic form captures parity of overlapping implications, which may correlate with bottleneck variables in resolution. This signature acts as a surrogate for topological rigidity in the implication space.

## Novelty

- Judge: NOVEL over 12 arXiv hits

Top hits consulted: - [2405.02292v1] ALOHA 2: An Enhanced Low-Cost Hardware for Bimanual Teleoperation - [1303.4033v1] Two-dimensional magnetic interactions in  $\text{LaFeAsO}$  - [9803030v1] Search for the Proton Decay Mode proton to neutrino  $K^+$  in Soudan 2 - [1911.05009v4] On Solvable Quadratic Lie algebras having an Abelian descending central ideal - [2004.07430v2] Ideals generated by  $a$ -fold products of linear forms have linear graded free resolution

## Empirical Test

- exit code: 0, elapsed: 33.30s

```
|w-|sig||=3
Testing n=4, m=1
Testing n=4, m=2
Testing n=4, m=3
Testing n=4, m=4
n=4, m=4, w=5, sig=0, |w-|sig||=5
n=4, m=4, w=5, sig=0, |w-|sig||=5
n=4, m=4, w=5, sig=0, |w-|sig||=5
Testing n=4, m=5
n=4, m=5, w=5, sig=0, |w-|sig||=5
n=4, m=5, w=5, sig=0, |w-|sig||=5
n=4, m=5, w=5, sig=0, |w-|sig||=5
Testing n=4, m=6
n=4, m=6, w=4, sig=0, |w-|sig||=4
n=4, m=6, w=5, sig=0, |w-|sig||=5
n=4, m=6, w=5, sig=1, |w-|sig||=4
Testing n=4, m=7
n=4, m=7, w=5, sig=0, |w-|sig||=5
n=4, m=7, w=5, sig=0, |w-|sig||=5
n=4, m=7, w=4, sig=0, |w-|sig||=4
n=4, m=7, w=5, sig=0, |w-|sig||=5
n=4, m=7, w=5, sig=0, |w-|sig||=5
Testing n=4, m=8
n=4, m=8, w=5, sig=0, |w-|sig||=5
n=4, m=8, w=5, sig=0, |w-|sig||=5
n=4, m=8, w=5, sig=0, |w-|sig||=5
n=4, m=8, w=5, sig=0, |w-|sig||=5
n=4, m=8, w=5, sig=0, |w-|sig||=5
Testing n=5, m=1
```

```

Testing n=5, m=2
Testing n=5, m=3
Testing n=5, m=4
Testing n=5, m=5
n=5, m=5, w=5, sig=0, |w-|sig||=5
n=5, m=5, w=6, sig=0, |w-|sig||=6
Testing n=5, m=6
n=5, m=6, w=6, sig=0, |w-|sig||=6
n=5, m=6, w=6, sig=0, |w-|sig||=6
Testing n=5, m=7
n=5, m=7, w=6, sig=0, |w-|sig||=6
n=5, m=7, w=6, sig=0, |w-|sig||=6
Testing n=5, m=8
n=5, m=8, w=6, sig=0, |w-|sig||=6
n=5, m=8, w=6, sig=0, |w-|sig||=6
n=5, m=8, w=6, sig=0, |w-|sig||=6
n=5, m=8, w=6, sig=0, |w-|sig||=6
Testing n=5, m=9
n=5, m=9, w=6, sig=0, |w-|sig||=6
n=5, m=9, w=6, sig=0, |w-|sig||=6
Testing n=5, m=10
n=5, m=10, w=6, sig=0, |w-|sig||=6
n=5, m=10, w=6, sig=0, |w-|sig||=6
n=5, m=10, w=6, sig=0, |w-|sig||=6
Testing n=6, m=1
Testing n=6, m=2
Testing n=6, m=3
Testing n=6, m=4
Testing n=6, m=5
Testing n=6, m=6
Testing n=6, m=7
n=6, m=7, w=7, sig=0, |w-|sig||=7
Testing n=6, m=8
n=6, m=8, w=7, sig=0, |w-|sig||=7
Testing n=6, m=9
n=6, m=9, w=7, sig=0, |w-|sig||=7
Testing n=6, m=10
n=6, m=10, w=7, sig=0, |w-|sig||=7
n=6, m=10, w=7, sig=0, |w-|sig||=7
Testing n=6, m=11
n=6, m=11, w=7, sig=0, |w-|sig||=7
Testing n=6, m=12
n=6, m=12, w=7, sig=-1, |w-|sig||=6
n=6, m=12, w=7, sig=-1, |w-|sig||=6
RESULT: FALSIFIED n=3 m=3 w=3 sig=0

```

### Judge reasoning

The test found a counterexample with  $n=3$ ,  $m=3$ ,  $w=3$ , and  $\text{sig}=0$ , where  $|w-|\text{sig}|| = 3$ , which exceeds the conjectured bound of 2. | next: Investigate the specific case of  $n=3$ ,  $m=3$ , to understand why the conjecture fails and potentially refine the conjecture.

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## Convex Hull Facet Count and Resolution Proof Size

- **Verdict:** FALSIFIED

- **Bridge:** Convex geometry × Resolution proof size
- **Recorded:** 2026-04-24 13:08 UTC
- **Entry ID:** 98ce17e2db79

## Statement

The number of facets of the convex hull of the clauses in a 3-SAT instance is  $\Theta(2^{\{n/2\}})$  when the resolution proof size is  $\Theta(n)$ .

## Rationale

Convex geometry provides tools to analyze the structure of clause spaces, which may reveal geometric constraints on resolution proofs. The facet count could reflect the complexity of navigating the clause space during resolution.

## Novelty

- Judge: NOVEL over 0 arXiv hits

## Empirical Test

- exit code: 1, elapsed: 0.87s

```
TRIAL: {'metric_name': 'convex_hull_facets', 'metric_value': 7, 'instances_tested': 1, 'conjecture_h
TRIAL: {'metric_name': 'convex_hull_facets', 'metric_value': 7, 'instances_tested': 1, 'conjecture_h
TRIAL: {'metric_name': 'convex_hull_facets', 'metric_value': 2, 'instances_tested': 1, 'conjecture_h
TRIAL: {'metric_name': 'convex_hull_facets', 'metric_value': 3, 'instances_tested': 1, 'conjecture_h
TRIAL: {'metric_name': 'convex_hull_facets', 'metric_value': 1, 'instances_tested': 1, 'conjecture_h
```

--STDERR--

Traceback (most recent call last):

```
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_067316bd.py", line 58, in <module>
    first_failing_seed = next(r["seed"] for r in results if not r["conjecture_holds"])
                        ~~~~~^
File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_067316bd.py", line 58, in <genexpr>
    first_failing_seed = next(r["seed"] for r in results if not r["conjecture_holds"])
                        ~~~~~^
```

KeyError: 'seed'

## Judge reasoning

Multiple counterexamples show actual facet counts (e.g., 7, 2, 3) vastly below conjectured  $\Theta(2^{\{n/2\}})$  values (e.g., 128, 6, 16). | next: Analyze specific 3-SAT instances where convex hull facet counts deviate from  $\Theta(2^{\{n/2\}})$ , verifying resolution proof size scaling.